voltage_divider_routed

Design Report

Rev 1 | 2026-10-14 | jlcpcb

kicad-tools 0.22.0

Board Summary

Property	Value
Layers	2 copper (F.Cu, B.Cu)
Footprints	4 (2 SMD, 2 THT, 0 other)
Nets	3
Traces	14 segments
Vias	4
Board Size	30.0 x 25.0 mm

Design Overview

Theory of Operation

Voltage Divider Test

Simple 2-resistor voltage divider

5V -> 2.5V (10k/10k)

voltage_divider_routed

Power Architecture

Power Rails: GND, VIN

Hand-Solder (THT) Components

The following 2 through-hole components are **excluded from the SMT pick-and-place file** and must be hand-soldered (or wave/selective-soldered) after SMT assembly. They appear in the BOM for sourcing.

Value	Package	Qty	References
IN	PinHeader_1x02_P2.54 mm_Vertical	1	J1
OUT	PinHeader_1x02_P2.54 mm_Vertical	1	J2

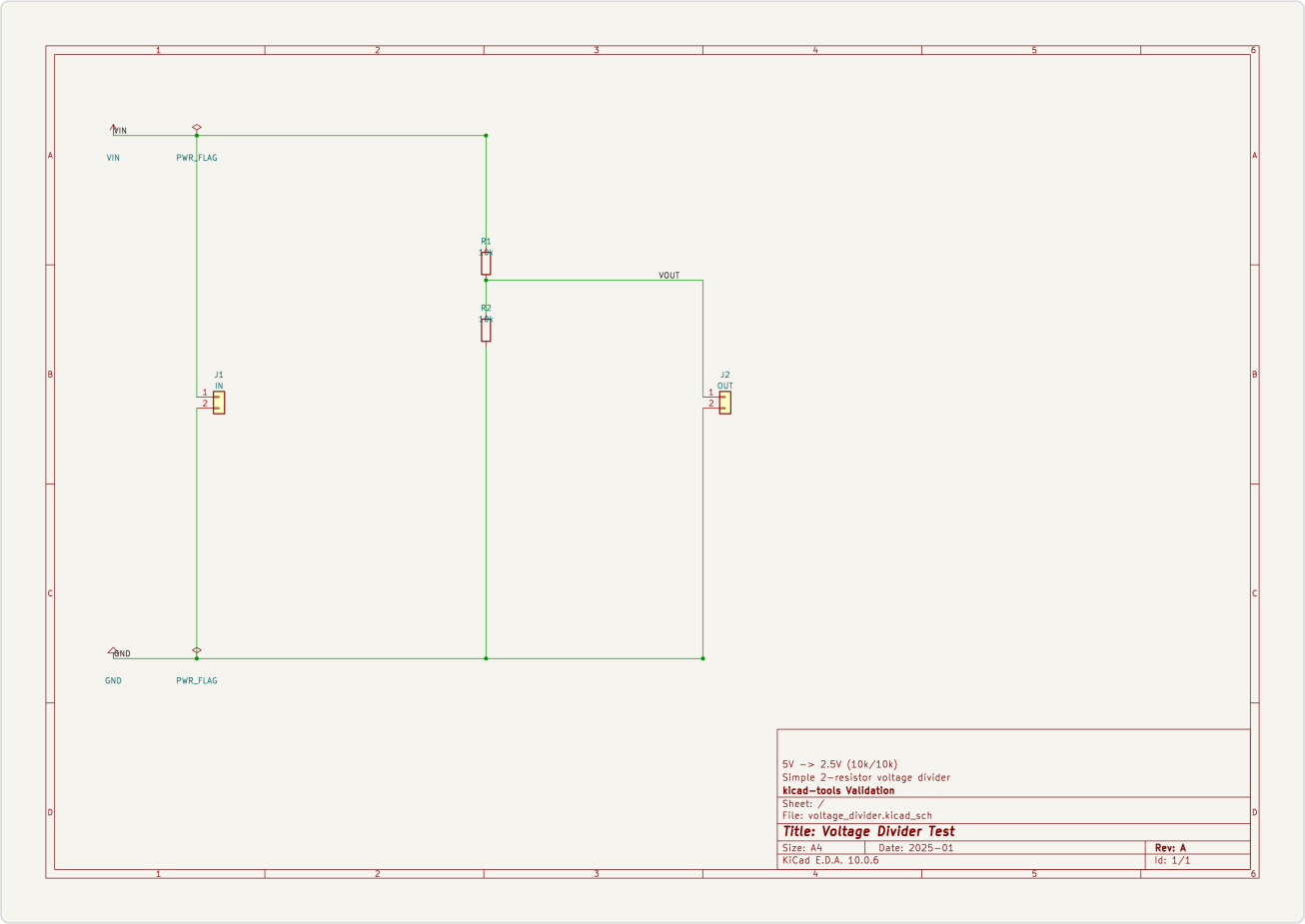
ERC Status

Metric	Count
Errors	0
Warnings	0

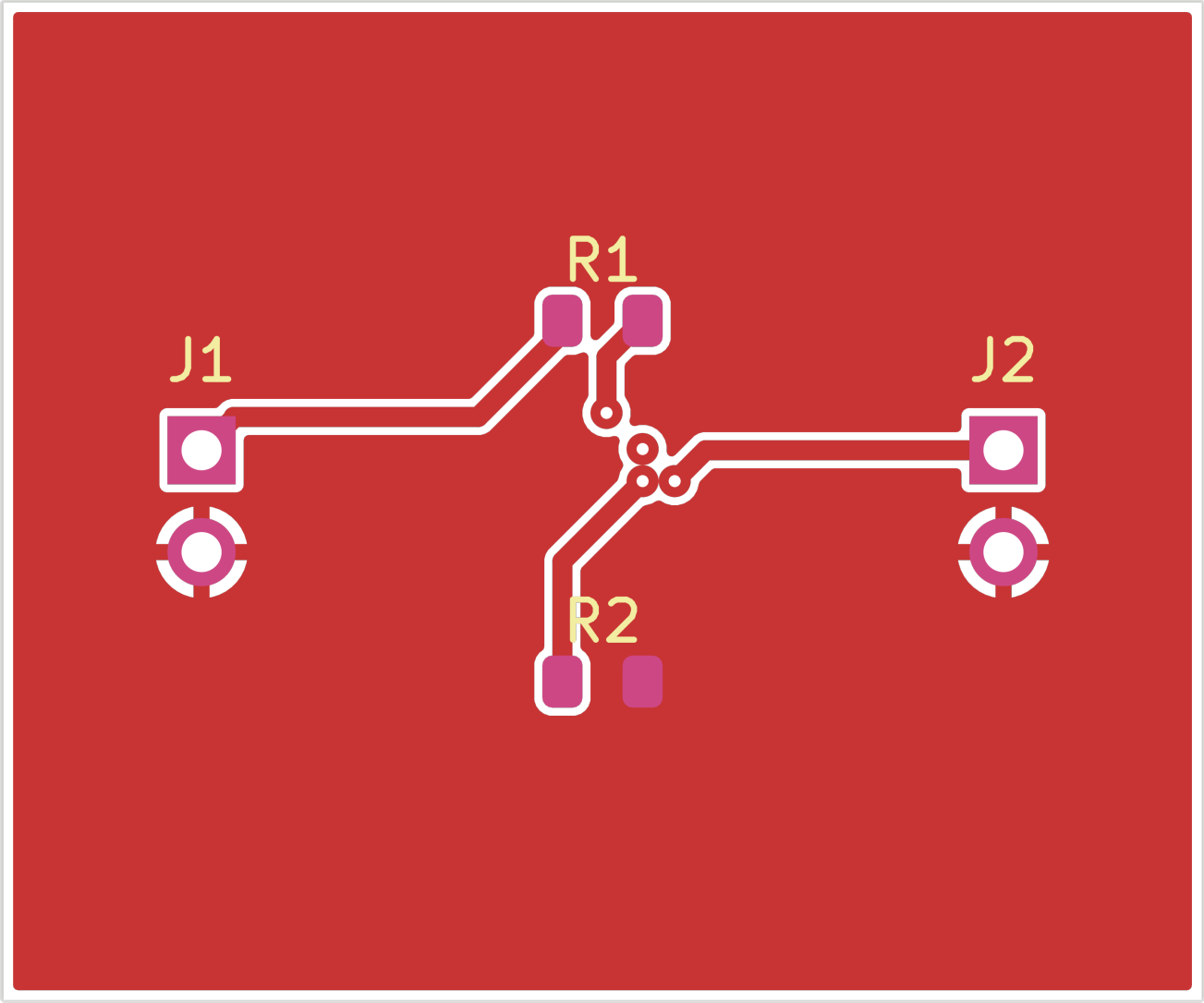
Status: SKIPPED -- ERC skipped by user request

Schematic Overview

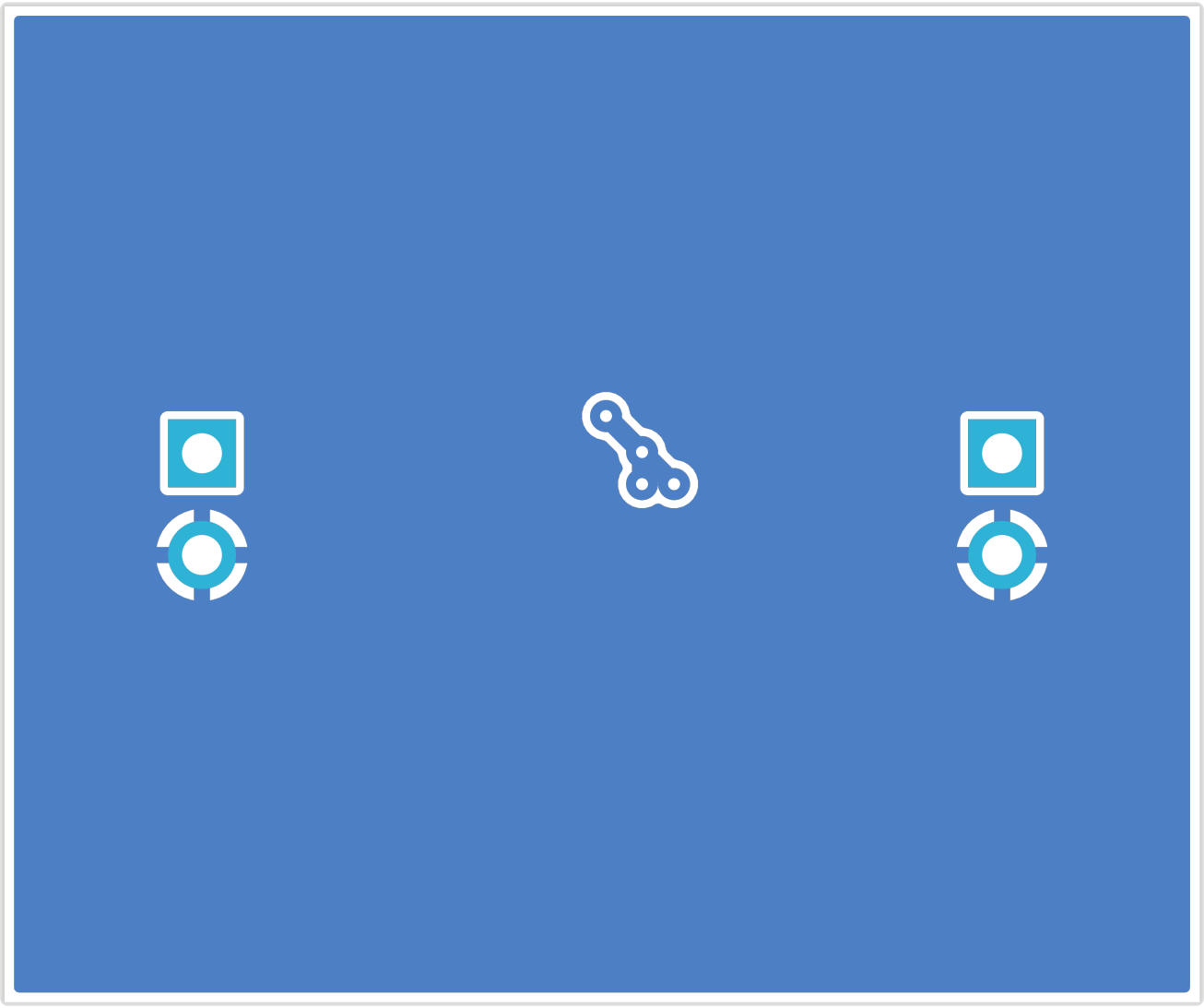
Schematic: voltage_divider



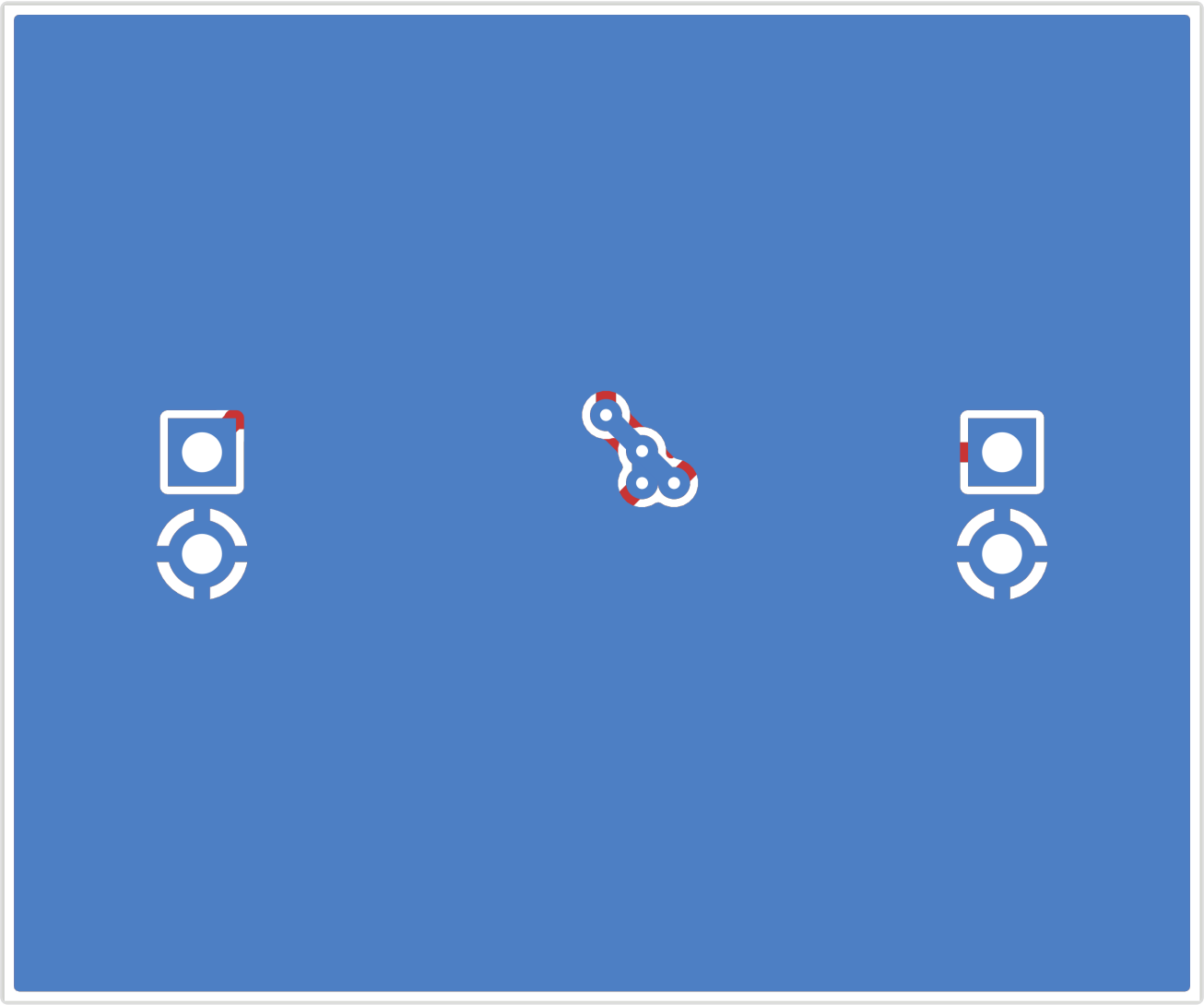
PCB Layout



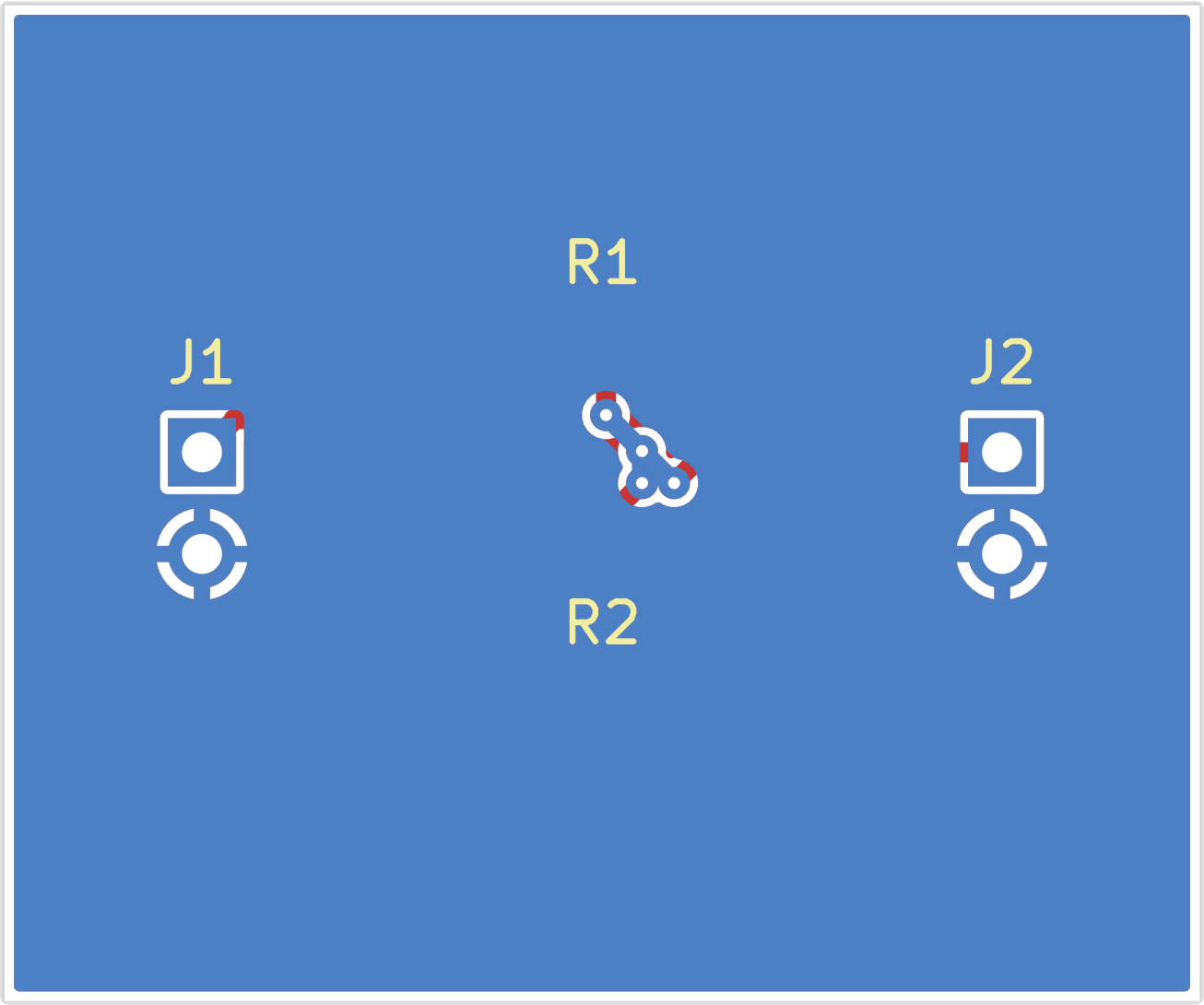
voltage_divider_routed



Copper

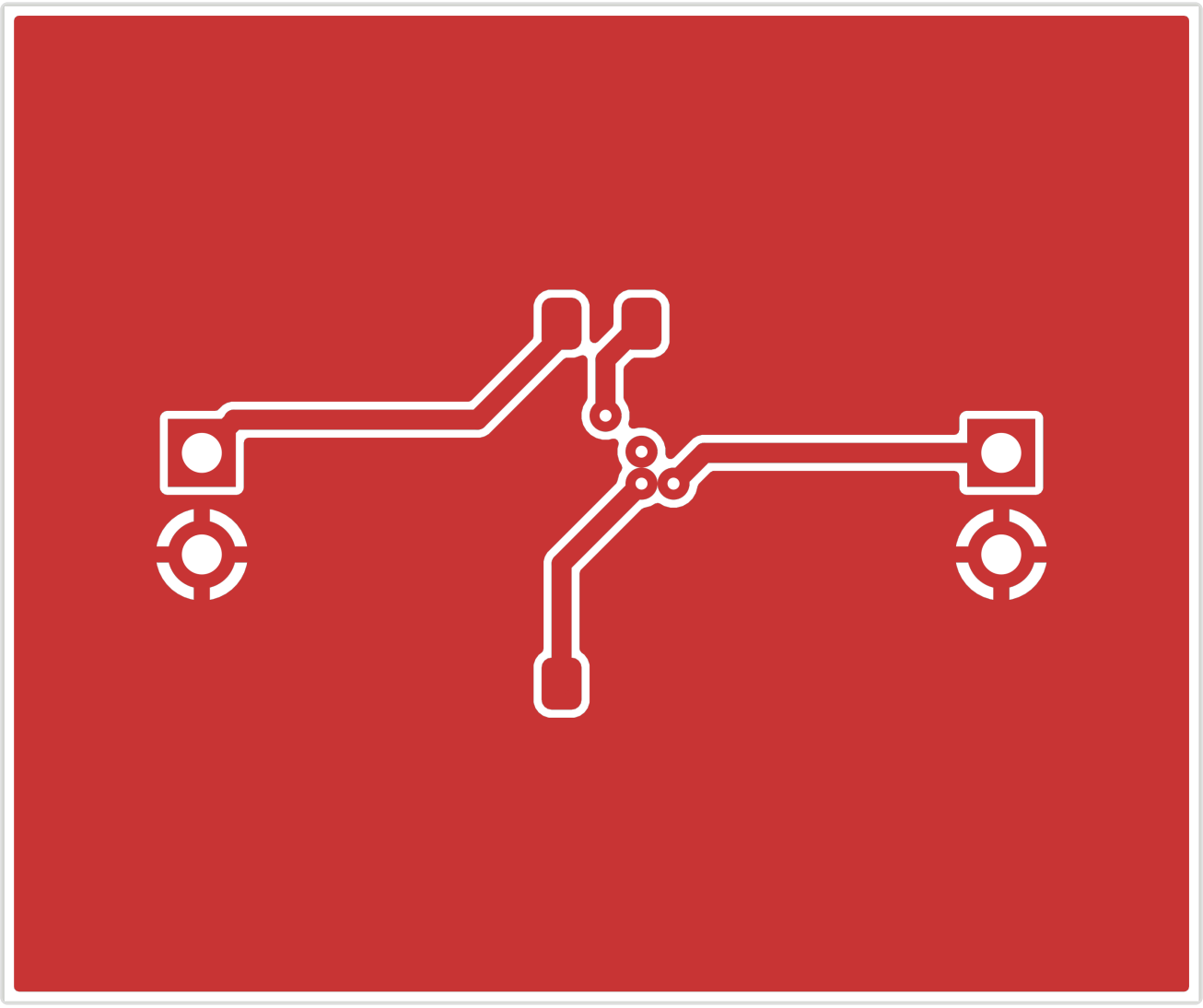


Assembly

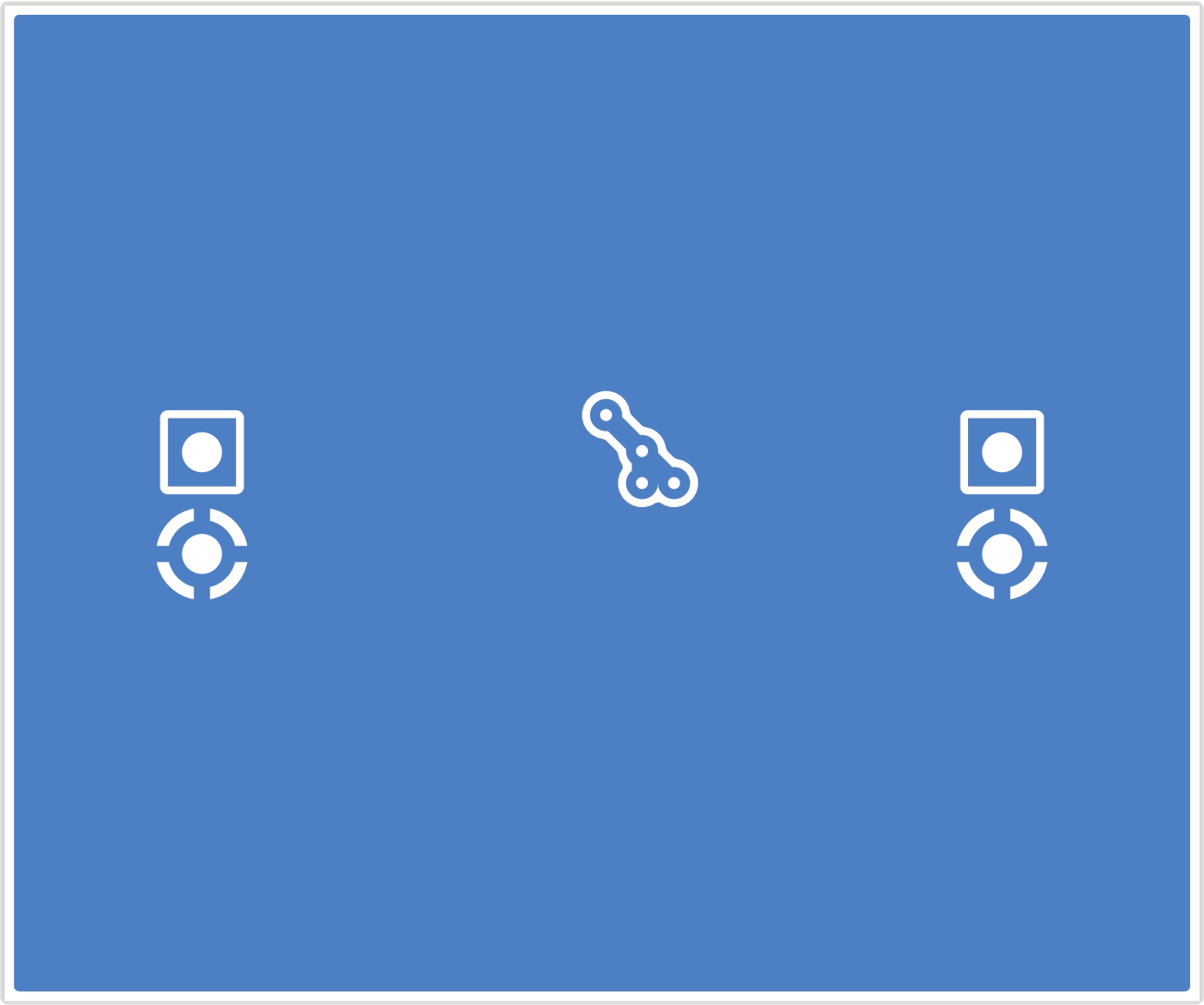


Copper Layers

F.Cu



B.Cu



Bill of Materials

Value	Package	Qty	References
IN	PinHeader_1x02_P2.54 mm_Vertical	1	J1
OUT	PinHeader_1x02_P2.54 mm_Vertical	1	J2
10k	R_0805_2012Metric	2	R1, R2

DRC Status

Metric	Count
Errors	0
Warnings	0
Blocking	0

Status: PASS

Manufacturing Readiness

Verdict: READY

Action Items

- **[OPTIONAL]** Verify zone fill in KiCad for 1 zone-connected nets

Routing Status

Metric	Value
Signal Net Completion	100.0% (2/2)
Overall Completion	100.0%
Complete Nets	3 / 3
Zone-Connected Nets	1
Incomplete Nets	0
Unconnected Pads	0

Zone-Connected Nets

- GND

Cost Estimate

Metric	Per Board (estimated)
PCB Fabrication	~0.55 USD
Components (estimated)	~0.21 USD
Assembly (estimated)	~1.93 USD
Total (estimated)	**~2.69 USD**
Batch Quantity	5
Batch Total (estimated)	~13.44 USD